

Federico Palatnik Moroz

Backend Software Engineer · API Integration · Automation Systems · Distributed Systems

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- ▶ github.com/federicomoroz
- ▶ federicomoroz.github.io
- ▶ English C2
- ▶ Remote · UTC-3, full overlap with US business hours

PROFESSIONAL SUMMARY

Backend Software Engineer building REST APIs, third-party platform integrations, asynchronous workflows, queue systems and distributed services in Python and Java. In my most recent role I cut operational processing time from approximately 10 minutes to 5 seconds across ~2,100 monthly transactions.

I integrate AI with deterministic code: guardrails, validation and caching layers keep LLM calls predictable and cost-efficient. Clean architecture, SOLID principles and real test coverage — the code is public and auditable. Full professional proficiency in English (C2), with a track record of delivering projects end to end in English. Based in Buenos Aires (UTC-3), which overlaps the entire US business day.

TECHNICAL SKILLS

- Areas:** Backend Development, Distributed Systems, Microservices, Queue Systems, Asynchronous Processing, Process Automation, Database Design, AI Integration, LLM Orchestration
- Languages:** Python, Java, JavaScript, TypeScript, C#, SQL, PHP
- Frameworks & Libraries:** FastAPI, Flask, Node.js, React, SQLAlchemy, Alembic, Pydantic v2, dependency-injector, httpx, Spring Boot, Spring Data JPA, Spring Kafka, Hibernate, Maven, Django, Laravel
- Databases & Infrastructure:** Apache Kafka, Redis, PostgreSQL, Flyway, SQLite, Qdrant (Vector DB), Docker, Docker Compose, Unicorn, APScheduler, Render
- AI & Integration:** Claude API (Anthropic), Voyage AI, Langfuse, RAG, Vector Search, Guardrails, LLM-as-Judge, Prompt Caching, Batch API, SSE Streaming, n8n, Airtable, BigQuery, REST APIs, Webhooks, SendGrid, Telegram Bot API
- Testing & Quality:** pytest, pytest-asyncio, pytest-cov, Playwright, mypy, Ruff, pre-commit, GitHub Actions, JUnit 5, ArchUnit, Testcontainers, architecture tests (AST and ArchUnit)
- Architecture & Practices:** OOP, SOLID, Clean Architecture, Hexagonal Architecture (Ports & Adapters), MVC, Repository Pattern, Composition Root, Event-Driven, Pipeline, Circuit Breaker, Transactional Outbox, Idempotent Consumer, Dependency Injection, Plugin Systems, CI/CD
- Development Tools:** Git, GitHub, Postman, VS Code

PROFESSIONAL EXPERIENCE

- Backend Developer** 2022 – Present
Freelance · International clients
- Designed and shipped production backend services in Python (FastAPI, Flask) and Java (Spring Boot): API gateways with rate limiting and circuit breakers, distributed task queues on Redis, notification routing engines, and Kafka-based microservices.
 - Built a chargeback resolution agent for a fintech use case: RAG with semantic search over Qdrant, deterministic guardrails, automated LLM-as-Judge evaluation, and n8n orchestration.
 - Developed a training plan generation platform on the Claude API, now in daily production use by a client.
 - Delivered API integrations and automation work to international clients through Fiverr, Upwork and direct referrals, working entirely in English.
- Backend Developer / Automation Engineer** Nov 2024 – Mar 2026
Del Mundo a Tus Manos · International e-commerce and logistics
- Integrated operations with third-party APIs — Amazon, Mercado Libre and Tienda Nube (LatAm e-commerce platforms), the national tax authority API, international couriers, and a Laravel/PHP backend — over REST, webhooks and purpose-built endpoints, handling authentication, path params, secrets management and IP allowlisting.
 - Cut operational processing time from approximately 10 minutes to 5 seconds across ~2,100 monthly transactions.

- Built operator-facing interfaces in Node.js, React and TypeScript on the Interface Extensions SDK, calling external endpoints client-side to avoid consuming the platform's automation-run quota.
- Diagnosed and fixed a silent failure on batches of 100+ records where requests hung with no trace in the execution log. Resolved with a 15 s AbortController, retry with exponential backoff, and non-blocking per-record error handling.
- Designed and maintained the operational data model — 11 relational tables, ~115,000 records and 48 automations in production — with global configuration centralized in a single parameters table instead of values hardcoded per automation.
- Built internal tooling: QR label generation and an AI-assisted WhatsApp inquiry routing system.

Game Developer and Programming Instructor

2021 – 2024

Freelance

- Built contract gameplay systems and modular frameworks in Unity and Unreal Engine: combat systems, AI controllers, and tooling for other developers.
- Taught C#, JavaScript, Python, C++ and game development to students across skill levels, with code reviews and project-based mentoring.

PROJECTS

CIRI Chargeback Agent

[github.com/federicomoroz/ciri-api-aux-sourcecode](#) · [ciri-chargeback-agent.onrender.com/panel](#) · [federicomoroz.github.io/en/projects/agente-contracargos/](#)

Agent that resolves fintech chargebacks: RAG with semantic search over Qdrant, deterministic guardrails —the code decides, the LLM explains— and n8n as an explicit orchestrator. It grades itself with an LLM judge, and a semantic cache cuts LLM cost by ~20%. High-risk cases stop and wait for a human analyst.

FastAPI · Claude API · Qdrant · Voyage AI · n8n · Langfuse · SQLite · 1232 Tests

AI Training Plan Generator

[federicomoroz.github.io/en/projects/mega-training-system/](#)

Web platform in daily production use by a client, generating training plans in real time on the Claude API. Plugin architecture: adding a discipline never touches the core. SSE streaming, a circuit breaker around the AI calls, dependency injection and an idempotency layer.

Flask 3.0 · Claude API · SSE · SQLite · Docker · dependency-injector · Pydantic v2 · 1704 Tests

Order Outbox Service

[github.com/federicomoroz/order-outbox-service](#) · [federicomoroz.github.io/en/projects/order-outbox-service/](#)

Two Java microservices solving the dual write problem: the order and its event are written in a single Postgres commit, and a relay publishes them to Kafka with exponential backoff. If the broker is down the order still succeeds. The consumer dedupes by event_id. The hexagonal rules are verified by ArchUnit: the build fails if a layer imports one it should not.

Java 21 · Spring Boot · Apache Kafka · PostgreSQL · Flyway · ArchUnit · Testcontainers · JUnit 5 · 72 unit tests

Eight more services, with source code and case studies published

[federicomoroz.github.io/en/projects/](#)

API gateway with sliding-window rate limiting and a Redis-backed circuit breaker (202 tests) · distributed task queue with scalable workers · shipping quoter with a hexagonal architecture traced end to end · multi-channel notification routing engine · webhook receiver · exchange rate monitor · environment validation CLI · turn-based combat framework in Unity.

EDUCATION AND LANGUAGES

Escuela Da Vinci — Programming — 2021 – 2023

UADE — Marketing — 2007 – 2010

Spanish — Native

English — Full Professional Proficiency (C2)